



Society of Petroleum Engineers

SPE-228936-MS

Innovative Water Shut-Off in Deep Water Fractured Carbonate Gas Reservoir Combining Production Logging Analysis With Zonal Isolation Using Swelling Polymer and Thixotropic Sealant Technologies for Highly Deviated Well

Fornari Maria Vittoria, ENI, HQ, Milan, Italy; Allegretti Gianluca, ENI, HQ, Milan, Italy; Cesale Davide, ENI, HQ, Milan, Italy; Brambilla Ruggero, ENI, HQ, Milan, Italy; Carpineta Gabriele, ENI, HQ, Milan, Italy; Mangione Alessandro, ENI, HQ, Milan, Italy; Salvatore Spagnolo, ENI, HQ, Milan, Italy; Ahmed E. Ghonim, Halliburton, MENA Region, Dubai; Xavier Alarcon, Halliburton, Egypt Area; Julio Vasquez, Halliburton, Egypt Area

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/228936-MS](https://doi.org/10.2118/228936-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

Water production can significantly impact gas production and its economic viability, especially in reservoirs with active aquifers. Water Shut-Off (WSO) operations with the combined challenges of deep water, deviated well and highly fractured carbonate open hole completed through slotted liner would not typically show high success rates. Even the diagnostics with Production Logging (PLT) in such an Open Hole (OH) environment may represent a significant challenge.

This case study is for a deep water well in a field considered one of the biggest gas producers that produces from a cretaceous naturally fractured carbonate reservoir supported by a huge water aquifer drive. Water loading inside these wells restricts gas production. Diagnostics should be performed to locate the water entry points. WSO operations require various intervention techniques between Electrical Wire Line (EWL) for PLT acquisition, plug setting and perforations whilst Coiled Tubing (CT) for WSO fluids placement. Simulations of these interventions were critical to conduct the operations safely.

Several lab tests were conducted to validate different potential WSO materials and carrying fluid for this operation. Formation packing was proposed for pumping into such huge, interconnected fracture system to provide enough cushion to allow for proper placement of the sealant system that exhibits thixotropic properties designed to resist gravity slumping in horizontal and highly deviated wells. Pumping of WSO sealant was conducted using CT equipped with an Inflatable Cement Retainer (ICR) to allow for selectivity and dual pumping injection while treating the reservoir. The final stage was conducted by placement of cement sheath to conduct proper isolation.

The customized solution enabled the reduction of water production and isolation of the bottom water sources. Additionally, potential NPT, operational risks and costs were reduced compared to drilling a new wellbore or even side-tracking the original wellbore which could be troublesome in a depleted/fractured

reservoir. First implementation of this solution was carefully planned to cover all potential contingencies during the intervention being the first implemented intervention of its kind in such complex environment.

The customized WSO design with sealant and cement have brought new insights and lessons learned for such a novel operation and new of its kind in a deep-water activities. Enhancements in well performance were granted after testing the well. The success and novels captured indicate several opportunities for operating companies to push the limits for solving water production problems related to the development of major deep-water fields.

Introduction

Water production is one of the main challenging phenomena within Oil and Gas industry. This problem comes with other operational and engineering challenges to properly identify and diagnose the accurate source of produced water within the field. This paper addresses the workflow for diagnosing, designing then executing a Water Shut Off (WSO) operation in offshore environment in Mediterranean Sea. Moreover, addressing the main lessons learned for such solution and its operation for relevant future applications. The field under discussion is located in the deep water region within Egyptian field where water depth is more than 1400 m (Fig.1).

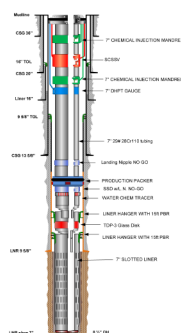


Figure 1—Well Completion Schematic

It's a gas reservoir producing from Cretaceous Natural Fractured Carbonate reservoir. This case study focus on a development well which was drilled throughout the reservoir where high density vugs and fractures exists. Upon production from this reservoir, depletion started to occur and water production started to increase over two years of dry gas production which led to having the problem of liquid loading within the well, affecting the gas production. The well was shut-in due to excessive water production and set for the analysis and investigation to find out the proper solution to be implemented (Fig.2).

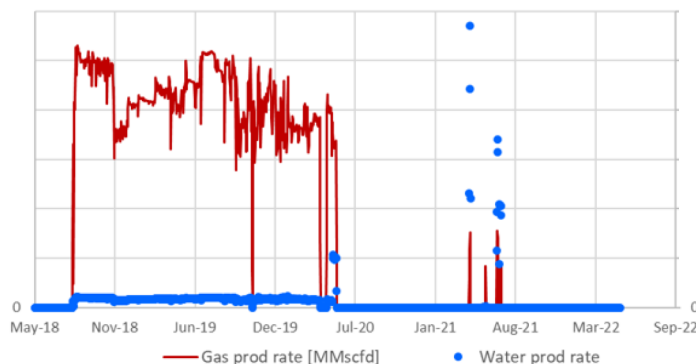


Figure 2—Gas & Water Production Trend since Production start

Case Challenge

The case encountered several engineering and operational challenges starting from the existing offshore environment to the challenging reservoir and completion nature. The well was a deviated development well with reservoir temperature (+/- 45 – 50 C) which is quite low due to the subsea well behavior. This is affecting the placement of thermal activated sealant. Also, the average reservoir pressure was around 4500 psi. In addition to the high Hydrogen Sulfide (H₂S) and Carbon Di-Oxide (CO₂) contents.

The first challenge within this case was to properly and accurately diagnose the new Gas-Water contact (GWC) required for a precise placement of the Water Shut Off (WSO) treatment without affecting the gas production from other intervals within pay zone.

Upon the initial analysis on the case, a trending coning and high permeable streaks behaviors are existing that are matching the presence of high density fractures within the reservoir.

Coiled Tubing (CT) intervention in subsea well through floating drilling rig was very challenging in several aspects. These challenges include the requirement of special designed CT string to allow safe deployment of inflatable elements and place the required treatments, while being within the subsea shearing limits and wellbore geometry and hydraulics limitations. Moreover, a optimized bottom-hole assemblies (BHAs) were utilized to enable more accurate and safe intervention and placement of WSO treatment stages without jeopardizing well safety. Inflatable Cement Retainer (ICR) was utilized to facilitate the injection and placement of WSO materials into water producing intervals while protecting the upper gas producing intervals.

Methodology and Operational sequence

Prior to implementation of the solution, several offline preparations were done to achieve the target objective starting from well re-entry through floating drilling rig. These steps and operations played fundamental role in the success of the well re-entry process and reduce any potential non-productive time (NPT) during the whole process.

After accessing the well and ensuring a clean wellbore, Electrical Wireline (EWL) was utilized to acquire PLT which analysis lead to the identification of the actual water producing intervals and gas producing intervals. Such acquisition is very essential to determine the actual Gas-Water contact (GWC) and final top of WSO treatment. The quality of the data was good, identifying the water sources and gas productive intervals. The water entry points were clearly detected with the gas hold up optical sensors incorporated within PLT tool ([Fig.3](#)). Further matching was performed with the open hole logs allowed the interpretation that water or fluid production is highly associated with the enhanced porosity intervals due to the diagenesis phenomena and open fractures.

The next stage post to PLT was the deployment of 7" bridge plug (BP) set by EWL at a depth of 10m below the top of water entry point identified by PLT. The purpose of this bridge plug is to reduce the rat-hole and avoid any excessive usage of WSO materials and precisely isolate the water producing interval only forcing the treatment into annulus of slotted liner. Subsequently, the slotted liner was perforated with e-line guns to allow the injection of swelling particles without risk of early screen out over the small slotted liner openings, allowing the treatment to reach and fill-up the reservoir fractures. The perforations depth was selected to be at depth of approximately 20m above the BP depth.

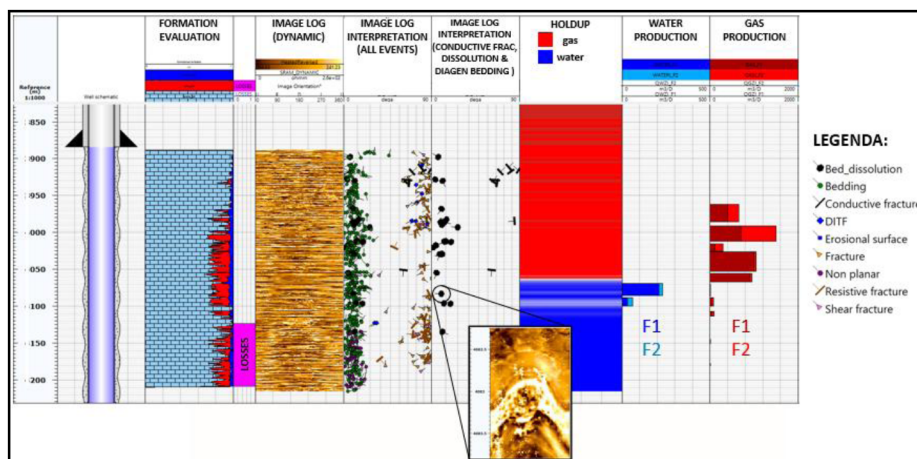


Figure 3—PLT result

Upon the successful completion of EWL operations, the well was displaced with Base Oil and Coiled Tubing (CT) was deployed with a Bottom-hole assembly (BHA) ending with open end nozzle to allow the placement of sorted crystalseal particle sizes of swelling polymer. CT was run in hole till tag the BP for correlating CT depths, then positioned in front of the perforated section to start injecting swelling particles stages. Three stages of low concentration slurry with swelling particles were pumped using Base Oil as carrier fluid and followed by fresh water as activation fluid. Those stages were pumped using the dual injection technique, pumping the treatment product inside CT and Base Oil in the annulus CT-tubing at the same time, to push inside formation all the swelling polymers.

At the end of the first WSO part, CT was POOH to change the BHA to include an Inflatable Cement Retainer (ICR) (Fig.4). Scope of ICR was to be set above the EWL perforations to force the injection of the second WSO fluid treatment to be placed through the newly added perforations. ICR helped to minimize the risk of having them being injected inside gas bearing zone and limited the movement of fluid across CT reducing potential CT pipe stuck. Once the ICR was set, the Thixotropic polymer sealant (TPS) was pumped downhole, displaced with Base Oil and allowed to be thickened. Then CT was released from ICR and pulled back to surface allowed the well to be in static conditions.

A third CT run was performed using an open end nozzle for placement of 10 barrels of cement plug above the ICR. After Waiting on Cement (WOC), CT was run to tag top of cement which was found 1 m over the top of the perforation area, confirming the sealing of the open-hole and slotted liner area and annulus.

At the end of WSO treatment, the well was opened to flow back to test the effectiveness of the treatment which revealed only a minimal reduction in water production.

BHA (schematic)	Item	Description	OD (in.)	ID (in.)	Length (ft.)	Connection	Fish Neck (in)	Serial Number												
	1	<u>EXTERNAL DIMPLE COIL CONNECTOR</u> BAKKE TYPE FOR 1.75" CT YIELD : 55,500 lbs MAX TEMP: 400 F MAKE UP TORQUE: 2,693 ft-lbs	2.875	1.375	0.4	2.375" PAC PIN	2.875													
	2	<u>MOTOR HEAD ASSEMBLY</u> BAKKE TYPE DISCONNECT BALL: 7/8" DISCONNECT SHEAR SCREWS 1000PSI x3 EACH SCREW CIRC SUB BALL: 3/4" CIRC SUB SHEAR SCREWS 900PSI x2 EACH SCREW BURST DISC: TBC MAKE UP TORQUE: 2,693 ft-lbs	2.875	0.67	2.75	2.375" PAC BOX x PIN	2.26 INT FN													
	3	<u>KNUCKLE JOINT</u> WORKING PRESSURE: 10,000PSI	2.875	1.125	1.04	2.375" PAC BOX x PIN	2.875													
	4	<u>WEIGHT BAR</u> WEIGHT: 16.4 KG MAKE UP TORQUE: 2,693 ft-lbs	2.875	1.78	3	2.375" PAC BOX x PIN	2.875													
	5	<u>KNUCKLE JOINT</u> WORKING PRESSURE: 10,000PSI	2.875	1.125	1.04	2.375" PAC BOX x PIN	2.875													
	6	<u>NON-ROTATING CENTRALIZER</u> MAKE UP TORQUE: 2,693 ft-lbs	TBC	0.75	1.44	2.375" PAC BOX x PIN	TBC													
	7	<u>CROSSOVER SUB</u> MAKE UP TORQUE: 2,693 ft-lbs	2.875	1	0.8	2.375 " PAC BOX x 2.375" WTS-8 PIN	2.875													
	8	<u>MECHANICAL DISCONNECT WITH DUAL FLAPPER</u> SHEAR RELEASE LOAD: 3 x 1200 lbs/SCREW	3.375	1	1.8	2.375" WTS-8 BOX x PIN	3.375													
	9	<u>3.375" INFLATABLE RETRIEVABLE PRODUCTION PACKER</u> <u>BASED ON CALCULATION</u> INFLATION RATIO: 2.6:1 DIFFERENTIAL PRESSURE: 6,080 psi BALL SIZE: 5/8" ANCHOR ORIENTATED: TOP ANCHOR MAX DIFF PRESSURE FROM ANCHOR SIDE: 1,470 psi SETTING PRESSURE : 2,100 psi DEFLATION FORCE: 8 x 1000 lbs MAX TEMPERATURE: 250 F	3.375	1.25	7.75	2.375" WTS-8 BOX	3.375													
<table><tr><td>Maximum O/D</td><td>➡</td><td>3.38 in</td><td>Overall BHA</td><td>➡</td><td>20.02</td></tr><tr><td>Minimum I/D</td><td>➡</td><td>0.67 in</td><td>Length (ft)</td><td></td><td></td></tr></table>									Maximum O/D	➡	3.38 in	Overall BHA	➡	20.02	Minimum I/D	➡	0.67 in	Length (ft)		
Maximum O/D	➡	3.38 in	Overall BHA	➡	20.02															
Minimum I/D	➡	0.67 in	Length (ft)																	

Figure 4—CT BHA with Inflatable Cement Retainer

Consequently, decision was made to perform another stage of WSO treatment.

Another e-line run was performed to perforate 20ft above the previous cement plug.

In the fourth CT run, a second ICR was deployed, but it could not be properly set above perforations because of cement debris presence landed on ball seat preventing its correct setting. Nevertheless, the second batch of Thixotropic polymer sealant was pumped and displaced with Base Oil, relying on the double injection technique for the correct batch placement.

upon pulling out of the hole, the ICR was found attached to the BHA and damaged.

In the fifth CT run, a circulating head was deployed, tagging the top of thixotropic polymer sealant treatment and the second cement plug was pumped above it and displaced with Base Oil.

After WOC, the well was opened to flow, confirming the success of the combined WSO treatment and cement plug placement, resulting in an estimated 60% reduction in produced water.

Fig. 5 shows a summary of operational sequences that has been performed over this case study.

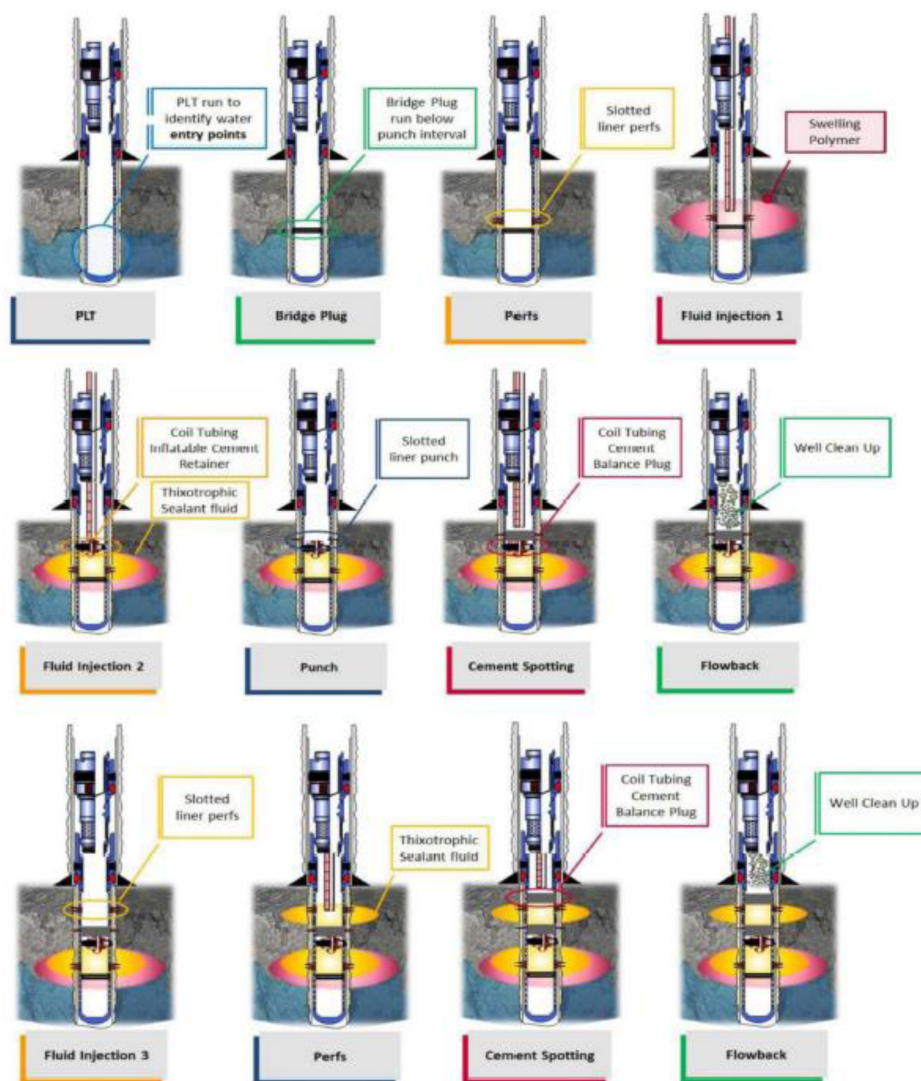


Figure 5—Summary of the WSO operations demonstrating the actual sequence

Water Swellable Polymer & Thixotropic Sealant Polymer Technologies

The Water Swellable Polymer (WSP) system is a crosslinked polyacrylamide synthetic polymer capable of absorbing 30 to 400 times its own weight in water. This material is available in different particle sizes from 40 microns up to 4 mm. Fig. 6 shows the physical appearance of the WSP (4-mm particle size) before and after it has been exposed to fresh water at 135°F for a few hours. The swelling rate can be delayed or accelerated, depending on the salinity of the carrier fluid, as Fig. 7 illustrates. The maximum rate of swelling is achieved with fresh water. CaCl₂ brines tend to slow down swelling even more than NaCl or KCl brines. No swelling is observed in hydrocarbon-based fluids. Temperature does not have any major effect on swelling properties. The stability of swollen WSP particles under H₂S and CO₂ environments was successfully evaluated. No degradation of the WSP was observed after 30 days of exposure to 1.2% H₂S and 7.5% CO₂ at 135°F in 65,000 ppm total dissolved solids (TDS) brine water.



Figure 6—Physical appearance of a 4-mm WSP before and after swelling in fresh water at 135°F (swelling factor greater than 400 after three hours of exposure to water).

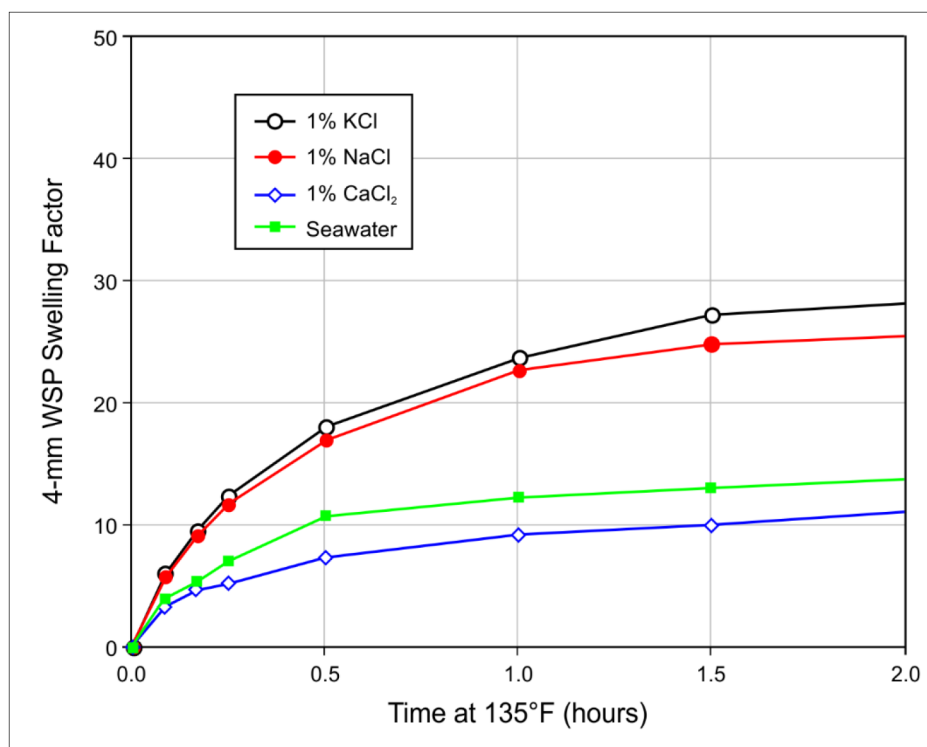


Figure 7—WSP swelling properties in different carrier fluids (4-mm particle size, 135°F).

The thixotropic conformance sealant (TCS) system is based on an organic-crosslinked polymer gel system combined with optimized rheological modifiers. It provides a predictable and controllable crosslinking time to obtain adequate placement. It can be placed into the wellbore by bullheading or using a coiled tubing (CT) unit. Once it is strategically placed into the zone to be shut off, in-situ gelation is activated by the reservoir temperature, plugging the permeability of the treated area, thereby limiting undesired fluid flow. The TCS system can be formulated to allow penetration into the formation matrix or to bridge off at the formation face for limited leak off, to help provide a controlled and shallow penetration at the formation face.

Conventional Organically Crosslinked Polymer (OCP) Gel System

The principle of application for the conventional OCP gel system is placement into the formation and propagation through the rock matrix, attempting to fill the pore spaces while displacing reservoir fluids. In-situ gelation occurs by means of a process activated by the well temperature, which plugs pore spaces and channels, thereby limiting undesired water or gas flow (Vasquez et al. 2005). At this point, a permanent barrier strategically placed only in the water (or gas) producing zone is formed, given that the hydrocarbon-producing zones can be mechanically or chemically protected.

Additionally, the conventional OCP system is a low-viscosity fluid system that can be easily injected deep into the matrix of the formation. The viscosity of the OCP system is approximately 20 to 30 cp at room temperature. The OCP system provides a predictable and controllable crosslinking time (phase change from liquid to gel state) to obtain adequate placement time for a wide temperature range. Fig. 8 illustrates the transition time from liquid to a 3D gel structure at predictable times. Gelation time corresponds to the inflection point of the viscosity versus time plot.

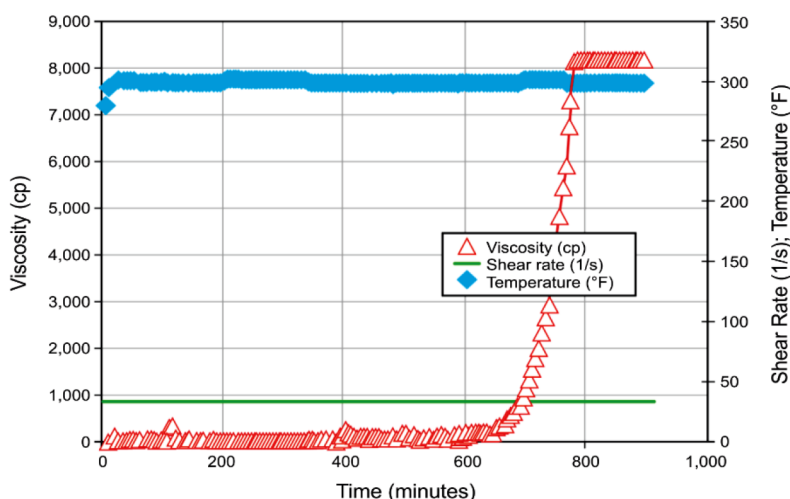


Figure 8—Typical gelation-time curve using a Brookfield high-pressure PVS viscometer (gelation time is approximately 13 hours at 300°F).

However, when these types of treatments are applied to horizontal or highly deviated wells, placement control becomes a critical factor. Depending on fluid density differences and wellbore deviation, the conventional OCP gel system may tend to slump/rise along horizontal sections, compromising the placement accuracy and overall success of the treatment.

TCS System Optimization

The rheological properties of the conventional OCP systems were improved by adding optimized rheological modifiers to minimize gravity slumping. The stress-dependent (thixotropic) rheological properties of the TCS system provides rapid viscosity increase after placement, allowing the treatment to remain in place until in-situ crosslinking occurs at a predicted time, helping provide a competent and precise seal across the targeted area. Once in place, the TCS system will crosslink at a predictable time, based on the bottomhole temperature in the near-wellbore region. Fig. 9 shows a comparison of the rheological properties between the conventional OCP system vs the improved TCS system.

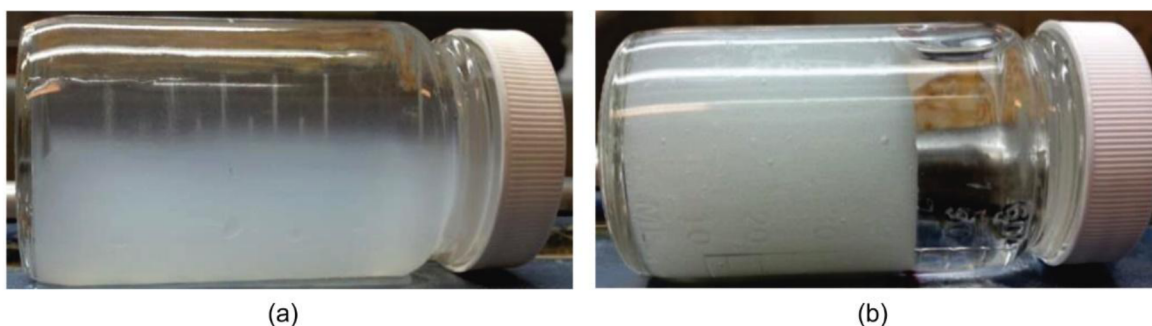


Figure 9—Bottle tests: (a) conventional OCP system slumps down along the side of the bottle, and (b) TCS system shows optimum placement in a horizontal scenario.

To highlight the concept of the optimized thixotropic properties in dynamic conditions, a laboratory model was assembled to mimic pumping the gel system into an annular space in a horizontal section (i.e. pumping the TCS system through CT in a horizontal well). In this model, clear plexiglass tubing was used, with an inner tube of 1/4-in. diameter fixed inside a tube of 1 1/2-in. diameter. The inner tube had 10 holes of approximately a 1-mm diameter approximately halfway down the length. The section of holes consisted of two holes placed approximately every 1/2-in. with radial symmetry and with the next set of holes perpendicular to the previous set. The entire assembly was filled with water and the gel systems were pumped into the inner tube and allowed to exit the holes and pass into the annular space. During both tests, the density of the polymer system was slightly higher than the water inside the tube. The first test used the conventional OCP system alone. As shown in Fig. 10.a, the polymer system exited the inner tube and fell to the low side of the outer tube (because of the density difference and low viscosity) and did not fill the entire annular space. Fig. 10.b shows the optimized TCS system. In this case, the high viscosity at low shear conditions caused the material to essentially stay in place as it exited the inner tube and build upon itself to fill the annulus, even with the density difference between the fluids.

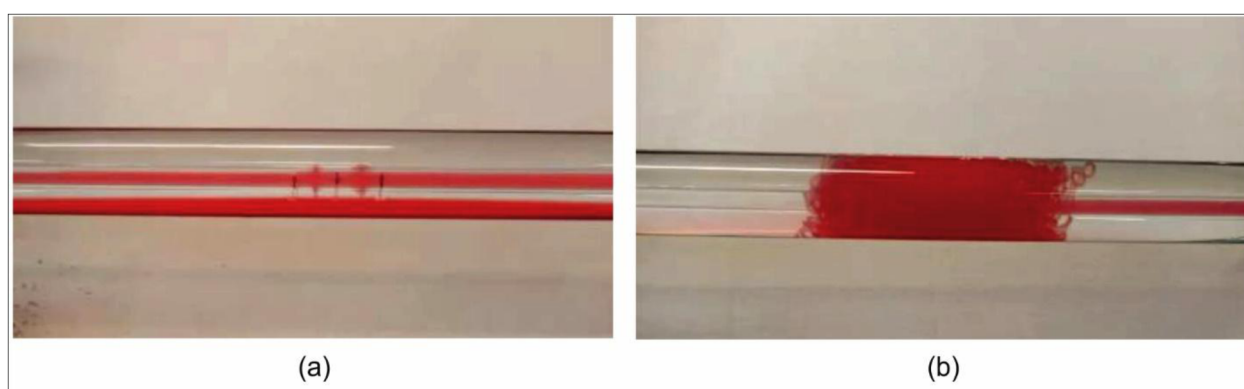


Figure 10—Dynamic placement test in a horizontal assembly: (a) conventional OCP system slumps down to the bottom section, and (b) TCS system shows optimum placement (both fluids were dyed red).

Additionally, it is important to highlight that the addition of the optimized rheological modifiers affect neither the sealing properties nor the thermal stability of the system. Gelation time is minimally changed. The TCS system provides the following benefits:

- Designed to prevent gravity slumping to form a chemical packer in horizontal and deviated wells.
- Applicable in high-angle/horizontal wells.
- Pumpable through CT.

- Can high withstand differential pressures. Tested in the laboratory up to 2,600 psi differential pressure. Controllable setting times.
- Permanent seal.
- Able to shut off water

Swellable polymer was utilized at different particle sizes to provide the required cushion and fill up the high density fractured system to regain matrix permeability and allow the sealant system to be placed properly. Such swelling polymer particles are usually pumped over a non-aqueous inert carrying fluid that doesn't allow the activation/swelling of the particles. Then, activating fluid is pumped to allow for swelling of such particles. Fig. 11 and Fig. 12 shows several lab testing trials were performed to precisely select the most appropriate fluid and assess its suspension capabilities for these particles.

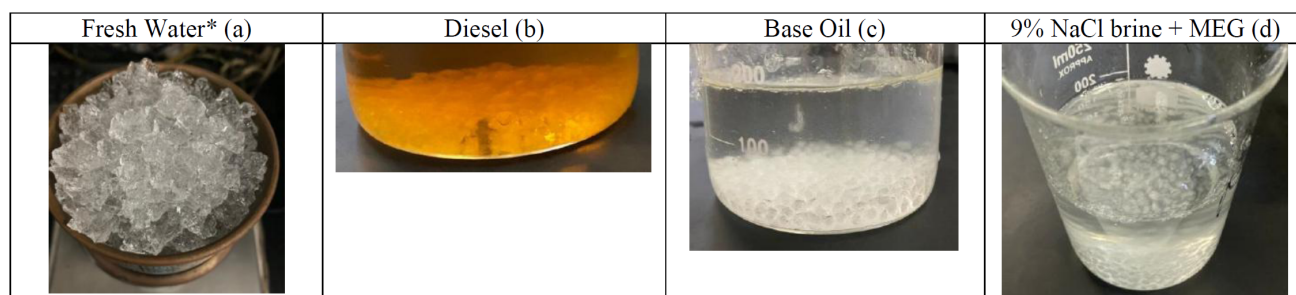


Figure 11—WSP swelling properties in different carrier fluids after 45 minutes: (a) Fresh Water, (b) Diesel, (c) Base Oil, and (d) 9% NaCl brine + MEG; Full swelling was observed in 30 minutes in Fresh Water.

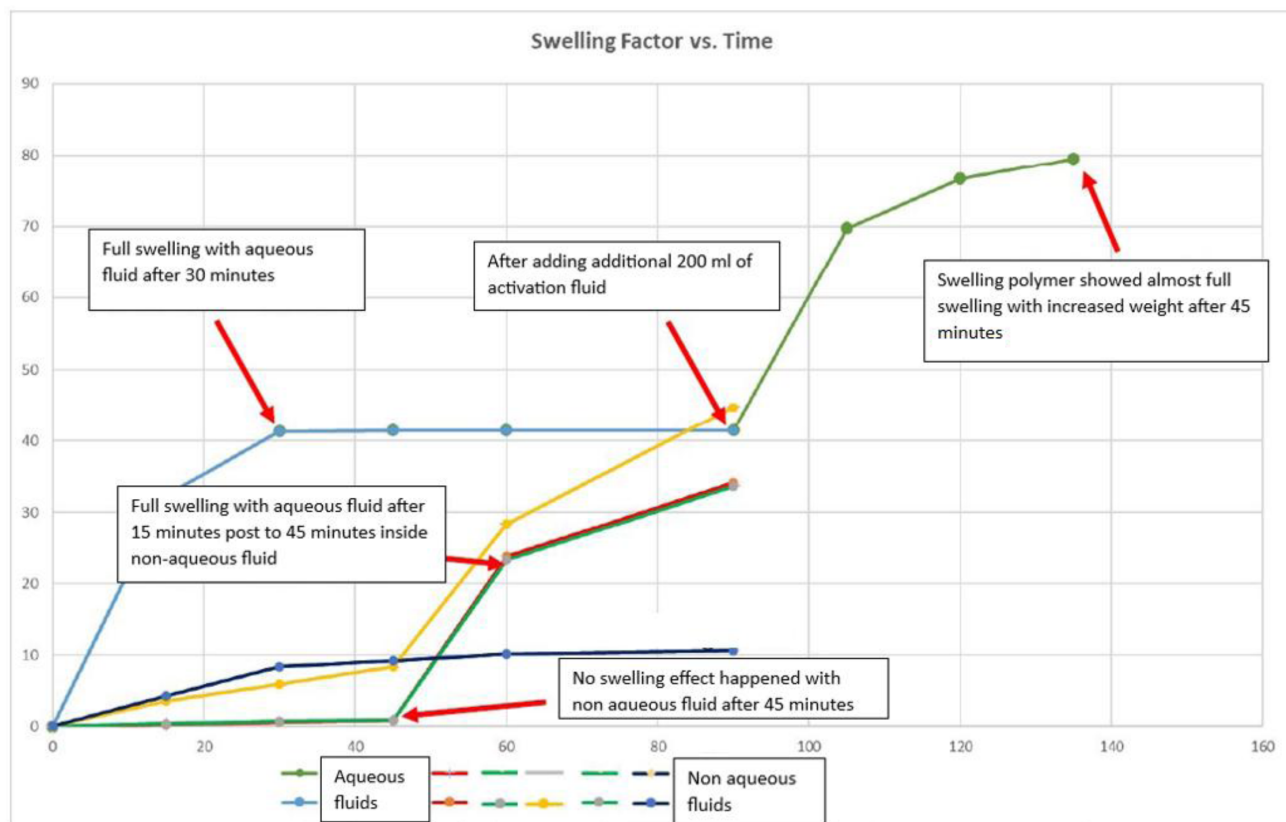


Figure 12—Swelling factor for different tested fluids vs time

Four potential carrier fluids for swellable polymers were tested: fresh water, diesel, base oil and 9% NaCl brine + MEG to simulate the pumping conditions. Here below is shown the condition of the swelling polymers after 45 minutes of contact.

The Thixotropic polymer sealant (TPS) system is an enhanced form of the conventional Organic Crosslinked Polymer (OCP), which was introduced in [Vasquez et al. 2005](#). The OCP system was used for more than 1000 operations across the world. However, for highly deviated and horizontal wells, slumping effects were observed. Therefore, additional two rheological modifiers were added to the blend to provide the required thixotropic properties required for this application. This allowed the system to be having the following added value features:

- Being, thermally activated that's controllable and predictable with the ability or easily clean out ([Vasquez et al. 2005](#) and [Al-Houti et al. 2017](#))
- Prevent gravity slumping effects within the highly deviated or horizontal wells.
- Thermal stability that's tested up to 250 F (40 C), validating the ability to sustain downhole conditions within case.
- Designing the treatment to be covering both required radial penetration and annular space between open hole section and slotted liner.

Lab tests were done to ensure the safe placement and holding of the TPS material over the placed cushion of crystalized swelling polymers.

[Fig. 13](#), shows the final simulation of both chemical solutions provided for this application.

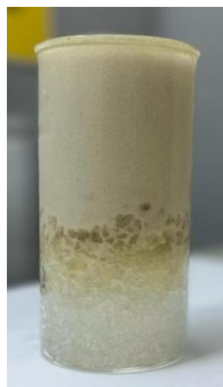


Figure 13—Final blended Thixotropic Sealant placed over fully swelled polymer particles in the lab, simulating the final placement blend/behavior inside reservoir

Observations and Conclusions

This application was the first of its kind worldwide to be performed in deep water/subsea environment combining all these applications for WSO treatment over NFC gas reservoir. Although, results were very promising, several observations and lessons learned were captured from this operation, which include the following:

- Complete diagnostics were very essential for production optimization and enable the accurate WSO applications in deep water or offshore environments.
- Higher volume of TPS is recommended to be pumped for similar scenarios to allow for more deeper penetration and coverage around wellbore specially with such high-density fractured reservoir.
- Swelling polymers concentration to be optimized and pumped using bigger working string or CT to avoid excessive usage of hydrocarbon-based fluids that are very challenging in supply in offshore

environments. In addition, since swelling polymers used for the first stage of WSO operation were water activated, no water based fluids could be used for the first stage of the WSO.

Results

The customized intervention solutions achieved water production reduction by 80% after two months of production compared to water production before WSO treatment. Such promising results created new opportunity for WSO applications in deep water and subsea applications within Oil and Gas industry.

References

1. M. V. Fornari, et al.: "Challenges and Innovations in Water Shut off for Naturally Fractured Reservoir in Subsea Wells", OTC-35075-MS, presented OTC 2024, Houston, TX, USA.
2. K. Matar, et al.: "World's Largest Yard Testing of a Thixotropic Polymer Sealant for Chemical Annular Plug Applications in Low-temperature Horizontal Wellbores", SPE-198085-MS, presented in SPE Kuwait Oil & Gas conference, 13-16 October 2019, Kuwait.
3. Al-Houti, N., Al-Othman, M., Al-Qassar, K., Al-Ebrahim, A., Matar, K., and Al Hamad, A. 2017. An Alternative Method for Cement Squeeze in North Kuwait: Case Study. Presented at the SPE/ICOTA Coiled Tubing and Well Intervention Conference and Exhibition, Houston, Texas, USA, 21-22 March. SPE-184802-MS. <https://doi.org/10.2118/184802-MS>.
4. Vasquez, J., Dalrymple, D., Eoff, L., Reddy, B., and Civan, F. 2005. Development and Evaluation of High-Temperature Conformance Polymer Systems. Presented at the International Symposium on Oil Field Chemistry, Houston, Texas, USA, 2-4 February. SPE-93156-MS. <http://dx.doi.org/10.2118/93156-MS>.